

Estimate the size of answers

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Estimate $6,782 \times 49$. Copy or complete the first step.

2. Estimate $92,410 \div 31$.

3. Estimate $487,205 + 218,770 - 96,420$.

4. Miro says: Miro rounds every number to one significant digit even when this gives a poor estimate. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate $6,782 \times 49$. Copy or complete the first step.
Answer About $6,800 \times 50 = 340,000$.
2. Estimate $92,410 \div 31$.
Answer About $90,000 \div 30 = 3,000$.
3. Estimate $487,205 + 218,770 - 96,420$.
Answer About $490,000 + 220,000 - 100,000 = 610,000$.
4. Miro says: Miro rounds every number to one significant digit even when this gives a poor estimate. Is this correct? Explain.
Answer Choose compatible rounded values that keep the calculation easy and reasonably close.

Estimate the size of answers

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Estimate $92,410 \div 31$.

2. Estimate $487,205 + 218,770 - 96,420$.

3. A calculator shows $7,204 \times 58 = 41,783$. Explain without exact calculation why it is wrong.

4. Identify and correct this misconception: Miro rounds every number to one significant digit even when this gives a poor estimate.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Estimate $92,410 / 31$.

Answer About $90,000 / 30 = 3,000$.

2. Estimate $487,205 + 218,770 - 96,420$.

Answer About $490,000 + 220,000 - 100,000 = 610,000$.

3. A calculator shows $7,204 \times 58 = 41,783$. Explain without exact calculation why it is wrong.

Answer The product should be about $7,200 \times 60 = 432,000$, so the displayed answer is far too small.

4. Identify and correct this misconception: Miro rounds every number to one significant digit even when this gives a poor estimate.

Answer Choose compatible rounded values that keep the calculation easy and reasonably close.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Estimate the size of answers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A calculator shows $7,204 \times 58 = 41,783$. Explain without exact calculation why it is wrong.

2. Explain why this reasoning fails: Miro rounds every number to one significant digit even when this gives a poor estimate.

3. Create a new example that tests the same idea as: Estimate $487,205 + 218,770 - 96,420$.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. A calculator shows $7,204 \times 58 = 41,783$. Explain without exact calculation why it is wrong.
Answer The product should be about $7,200 \times 60 = 432,000$, so the displayed answer is far too small.
2. Explain why this reasoning fails: Miro rounds every number to one significant digit even when this gives a poor estimate.
Answer Choose compatible rounded values that keep the calculation easy and reasonably close.
3. Create a new example that tests the same idea as: Estimate $487,205 + 218,770 - 96,420$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.